

CPSC 330 Lect

Feature impor

Announcements

- Midterm 1 grades are released on Pr
- Please remember to view your midte

Midterm Debrief

How did you feel about the first midterm?

- A. I felt well-prepared and it went smoothly
- B. I think it went okay. We'll see when we get the results back
- C. I struggled and didn't feel fully prepared
- D. I noticed some gaps between what I studied and what appeared on the exam
- E. It was a stressful experience for me

Motivating Feature importance

Consider the following two scenarios:

Scenario 1: Which model would you pick?

Predicting whether a patient is likely to be readmitted to the hospital based on features such as age, blood pressure, and BMI. You have two models:

- LGBM which results in 0.9 f1 score
- Logistic regression which results in 0.85 f1 score

Which model would you pick? Why?

Scenario 2

Predicting whether a user will purchase a product

based on their browsing history, previous purchases, and click behavior. You have two models:

- LGBM which results in 0.9 F1 score
- Logistic regression which results in 0.8 F1 score

Which model would you pick? Why?

Transparency

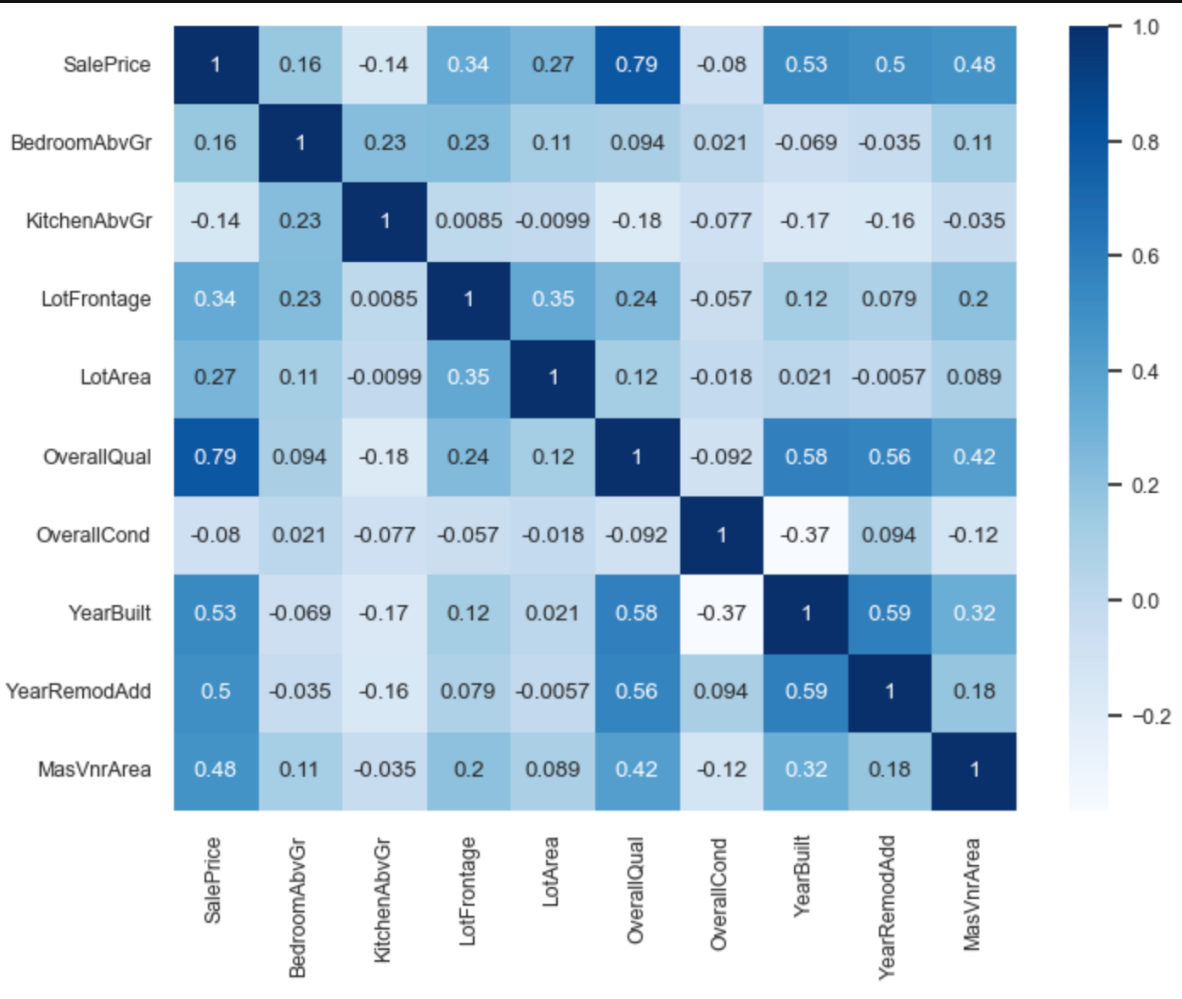
- In many domains understanding the **between features and predictions** and regulatory compliance.

Feature importances

- How does the output depend upon t
- How do the predictions change as a particular feature?

**How to get fear
importances?**

Correlations



- When limited correlation

Interpreting coefficient

- Linear models are interpretable because coefficients associated with different features represent the change in the target variable for a unit change in the feature's value, assuming all other features are held constant.
- Each coefficient represents the estimated change in the target variable for a unit change in the feature's value, assuming all other features are held constant.
- In a **Ridge** model,
 - A positive coefficient indicates that as the feature's value increases, the predicted value also increases.
 - A negative coefficient indicates that as the feature's value increases, the predicted value decreases.

Interpreting coefficients

- When we have different types of pre features, what challenges you might interpreting them?
 - Ordinally encoded features
 - One-hot encoded features
 - Scaled numeric features

Break

Let's take a break!



Pause and Reflect

We are now just over half-way through
You had a midterm already a couple of
like some feedback on how things are going
(the instructor).

Class Survey

I'd love to hear how you think lectures
how the course is going overall: bit.ly/c

Let's take a couple of minutes to comp
we get started on today's content.

Group Work: Class Demo

Coding

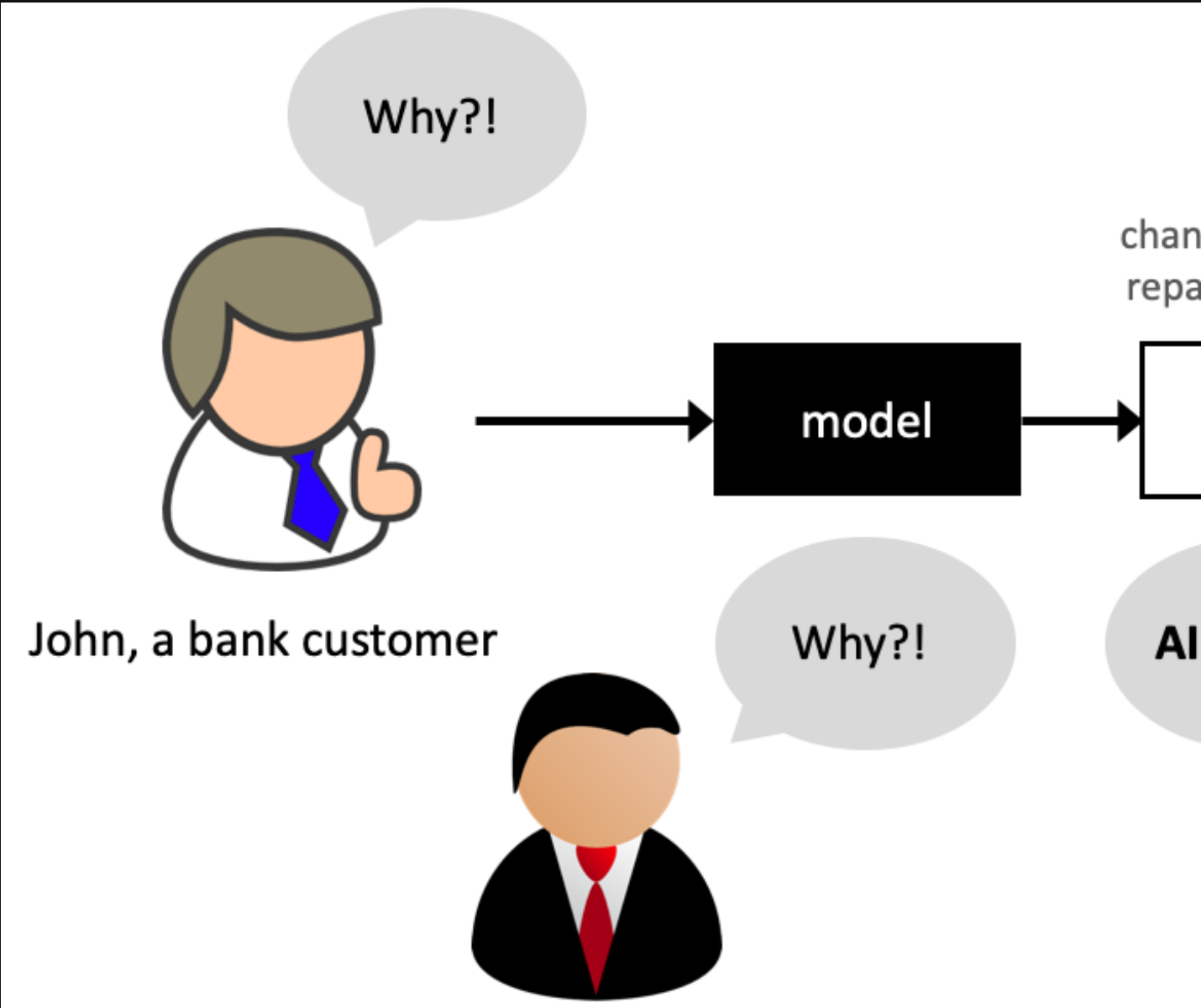
For this demo, each student should [click](#) create a new repo in their accounts, then clone the repo locally to follow along with the demo.

Group Work: Class Demo

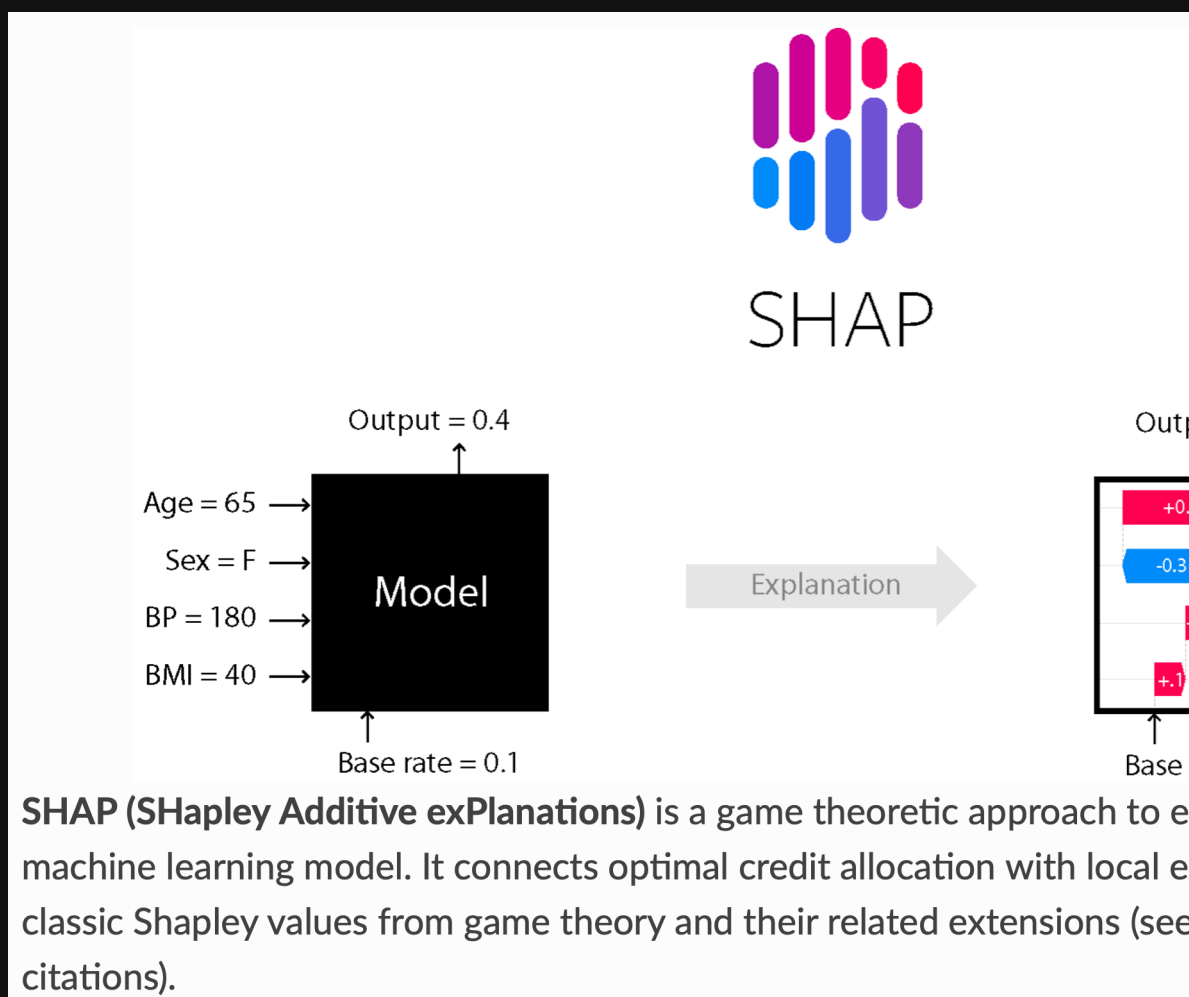
Coding (if time permits)

For this demo, each student should [click](#) create a new repo in their accounts, then clone the repo locally to follow along with the demo.

SHAP



SHAP



You might need to install **SHAP** in your environment

```
conda install -c conda-forge shap
```